AN EXPERIMENTAL ANALYSIS OF THE SPATIAL LOCATION HYPOTHESIS IN LEARNING

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In a recent study by Tolman, Ritichie, and Kalish (1) on spatial learning, an experiment was designed to test experimentally a revision of the senior author's "theory of expectancy." Animals were given four periods of preliminary training on apparatus presented in diagram in figure 1. The test trial was given on apparatus essentially the same as that presented in diagram in figure 2. (See discussion of apparatus in present article.)

The study by Tolman, Ritchie, and Kalish involves five assumptions relevant to our work:

1. The animals that chose path 6 (36 per cent) learned the location of the food box, and those that did not choose path 6 (64 per cent) did not learn the location of the food box.

2. The animals that chose path 6 acquired an expectation of food at L. "x expects food at location L."

3. Tolman, Ritchie, and Kalish assume that the 64 per cent which did not choose path 6 would have chosen path 6 had they been given training sufficient to determine the location of the food box. We quote:

> We believe that the reason that the remaining rats failed to take the short-cut was that they had not had enough training and thus had not yet learned the location of the food. With a few more days training we should have expected that the remaining rats would have chosen the short-cut (pp. 21–22).

4. The animals that chose path 6, the short-cut, inferred that it, of the 18 available pathways, was the pathway to location L.

5. They assume that the light during the training period became a cue of location L and that it was essential in the attainment of their results. We quote:

> We believe that such choices can only be made when there are distinctive stimuli in the environment which enable the rat to judge its own location relative to other places in the environment (p. 20). . . . We should say that they were running towards the location of the former goal, and that this location was indicated by the position of the light (p. 21).

Since the expression "x expects food at location L" is defined (p. 15) without reference to such distinguishable cues as the light employed in the training and in the test trial, we were led to wonder, among other things, whether the results reported demonstrated that (1) there were some x's that expected food at location L, or (2) whether the results obtained were not accountable for in terms of the

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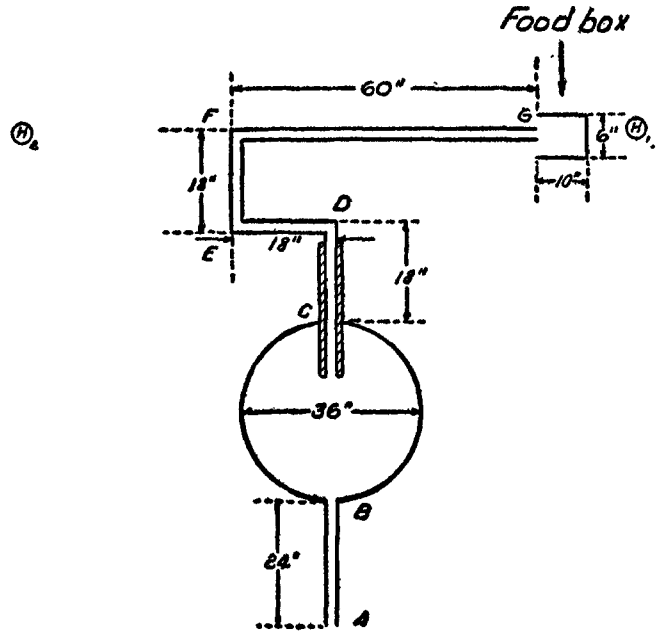


FIG. 1. APPARATUS USED IN PRELIMINARY TRAINING

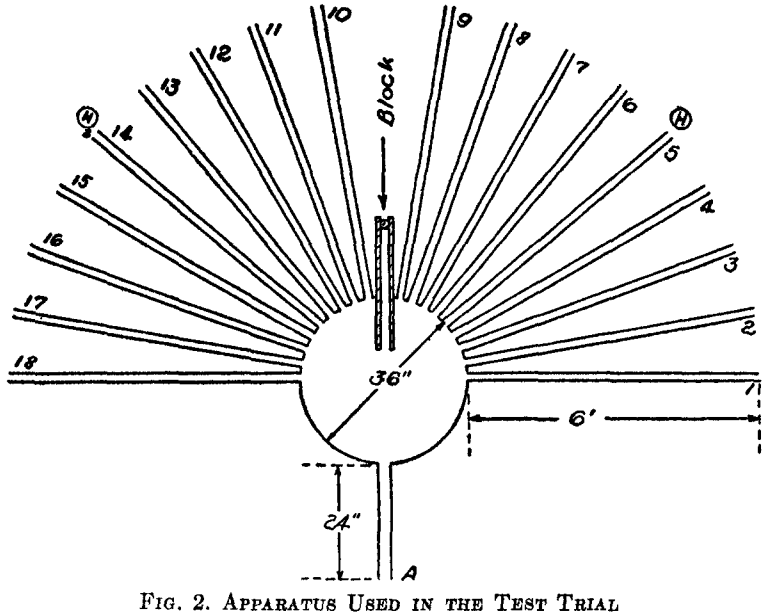


FIG. 2. APPARATUS USED IN THE TEST TRIAL

stimulus role of the light, or (3) whether these results were not explainable in other terms.

ANIMALS

Two hundred and forty-one animals were used in the present study. They were descendants of the Wistar strain reared in the Comparative Psychology Laboratory at the University of Texas and albinos purchased from a local dealer. The animals were between 90 and 150 days old at the beginning of the study. Fifty-seven animals of the group had had fourteen days of training two months previously on an H-type maze. All training was at night between seven and twelve P.M.

EXPERIMENTAL ROOM

The experimental room was 18 feet square with a window on the north, east, and west, and a door on the south. Three chairs and the experimental apparatus were the only equipment in the room during the experiment.

EXPERIMENT I

This experiment was designed to study the effects of the position of the light in the work of Tolman, Ritchie, and Kalish. They assert in their work that the animals (36 per cent) "were running toward the location of the former goal, and that this location was indicated by the position of the light" (p. 21). The present experiment studies the effects of different positions of the light upon choices made on the test trial.

Apparatus. The apparatus used was essentially the same as that used by Tolman, Ritchie, and Kalish. Figure 1 presents a diagram, with the dimensions listed, of the apparatus used in the preliminary training period. The food box used was 6″ x 6″ x 10″. A 7.5-watt bulb or 40-watt bulb in a desk lamp at H_1 or H_2 was the only illumination in the room.[2] In figure 2 we present a diagram of the apparatus used in the test trial. The light remained at the location on the test trial which it had occupied (H_1 or H_2) during preliminary training. The food box was removed on the test trial. On the test trial a block was placed in alley CD as shown in figure 2. Eighteen six-foot paths were spaced around the table on the test trial as shown in figure 2. In the work by Tolman, Ritchie, and Kalish paths thirteen through eighteen were only 24 inches long. The paths were so arranged that the table could be rotated independently of the paths.

Pre-test procedures. Two days before training began on the apparatus shown in figure 1, the animals were put on a 24-hour feeding schedule of Purina Dog Chow, being fed each evening at 8:30. Since our pre-test and test procedures were the same as those employed by Tolman, Ritchie, and Kalish, we quote from their work:

> On Day 1 the rats were given three trials. On the first trial they were put by hand into the food-box and allowed to eat for five min. On the second trial they were put in the middle of path FG and allowed to run to G and into the food-boxes. On the third trial, they were started at F and allowed to run into the food-boxes. They were then returned to their home cages and fed their full ration approximately 30 min. later.

[2] Tolman, Ritchie, and Kalish used a 5-watt bulb in their work. We could not obtain one from any local dealer at the time this study was made.

On Day 2 they were given three more trials. On the first trial they ran from F to the food-boxes. On the second trial they were put by hand into the alley on path CD and forced to run from there out onto path DE and from there to the food-boxes. This was repeated in the third trial.

On Day 3 they were again given three trials. On the first trial they were forced to run out of the alley on path CD. On the second and third trials they were started at A and allowed to explore the table-top, run through the tunnel and on to the food-boxes.

On Day 4 they were given three trials starting from A in the same manner as on the last two trials on Day 3. Thus, after their training on Day 4, each rat had run five times to the food-boxes at G, from the starting place at A.

Test procedures. On Day 5 one test trial was given. The apparatus was changed to that represented in fig. 2. Each rat was started at A, allowed to run into the blocked alley on CD, to return out of the alley, and to explore the table-top and the various alternative paths which radiated from it. The rat's trial ended as soon as it had chosen one of the paths and had run out to the end of it. If any rat took more than six min. to make such a choice it was removed. This was indicated in the protocol record by the expression "No choice." The circular table-top was revolved after each rat had run across it, and before the next one was started. This was done to prevent any possibility of "tracking" (Pp. 17–18).

Animals. Seventy-one animals between 120 and 150 days old were used in this experiment. All but twenty-seven had had prior training on an H-type maze two months previously. They were divided into three groups.

Group A was composed of 27 animals approximately 120 days old and had had no previous maze experience. The food box was at G and a 7.5-watt bulb was in a desk lamp at H_1 during preliminary training and on the test trial. Twenty-four animals made a choice on the test trial within six minutes.

Group B was composed of 22 animals approximately 150 days old and had had 14 days training on an H maze two months prior to this experiment. A 40-watt bulb was in a desk lamp at H_1 during the preliminary training and on the test trial. Twenty animals made a choice on the test trial.

Group C was composed of 22 rats which had had the same prior training as the animals in Group B. A 7.5-watt bulb was in a desk lamp at H_2 during preliminary training and on the test trial. Twenty-one made a choice on the test trial.

Results of experiment I. We present the results for groups A and C in figures 3 and 4 respectively in terms of the percentage of animals choosing each path on the test trial. The results of the work by Tolman, Ritchie, and Kalish are presented in figure 5. These data show that 36 per cent of the animals chose path 6 on the test trial. Our results for a repetition of this work, using a 7.5 watt light rather than a 5-watt light at H_1, are presented in figure 3. A comparison of the data presented in figures 3 and 5 shows that our results are at variance with the results of Tolman, Ritchie, and Kalish. Of our animals, 37.5 per cent chose paths 9 and 10, paths adjacent to blocked alley CD, and none chose path 6, "the path which points directly to location L." The data presented in figure 3 show that the dominant trend of choices was in the direction of path CD and not in the direction of the former goal which was indicated by the light at H_1 on the test trial. The data presented in figure 5 show that the choices for paths other than path 6 are nearly equally divided between paths 1–5 and 7–12. The

data presented in figure 3 show that 12.5 per cent chose paths 1–6 and 87.5 per cent paths 7–18.

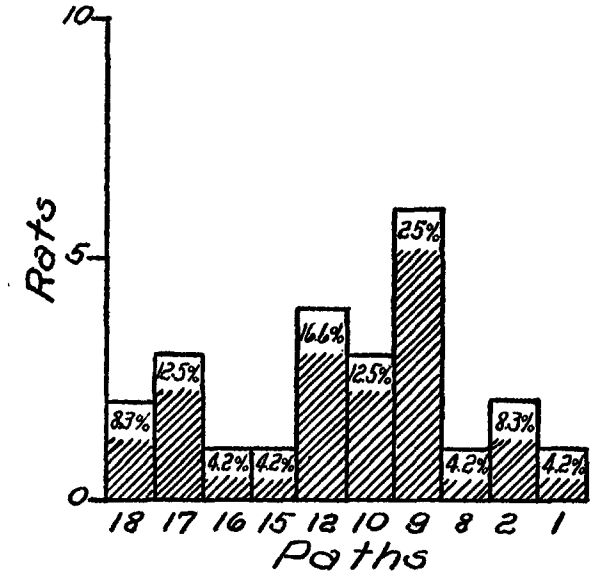


FIG. 3. NUMBER OF CHOICES ON TEST TRIAL; LIGHT AT H_1

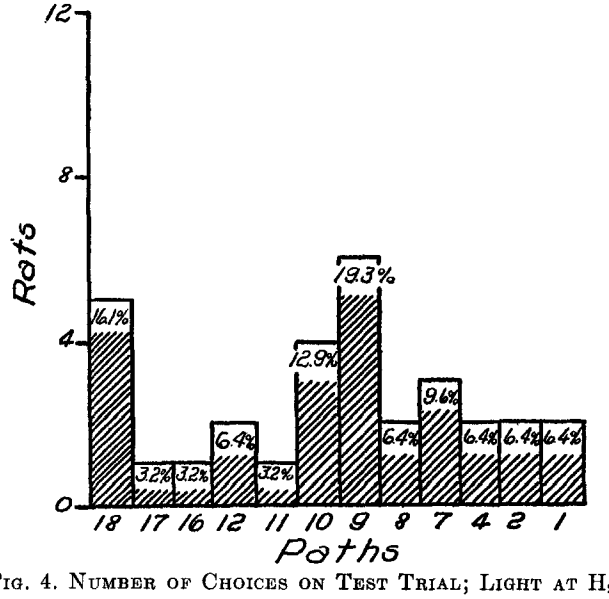


FIG. 4. NUMBER OF CHOICES ON TEST TRIAL; LIGHT AT H_2

We present the results of Group C in figure 4. These data show that 32.2 per cent of the animals chose paths 9 and 10 on the test trial when the light was at H_2 during preliminary training and on the test trial. This result is not essentially

different from the results of group A which had the light at H_1 during preliminary training and on the test trial.

Thirty-eight per cent of the animals of Group B chose paths 9 and 10, paths adjacent to blocked alley CD. The results obtained with a 40-watt bulb in a desk lamp at H_1 are substantially the same as the results of Group A (37.5 per cent) with a 7.5-watt bulb at H_1 and of Group C (32.2 per cent) with a 7.5-watt bulb in a desk lamp at H_2.

The results of this experiment indicate that neither the position of the light during preliminary training and on the test trial (Groups A and C), nor the

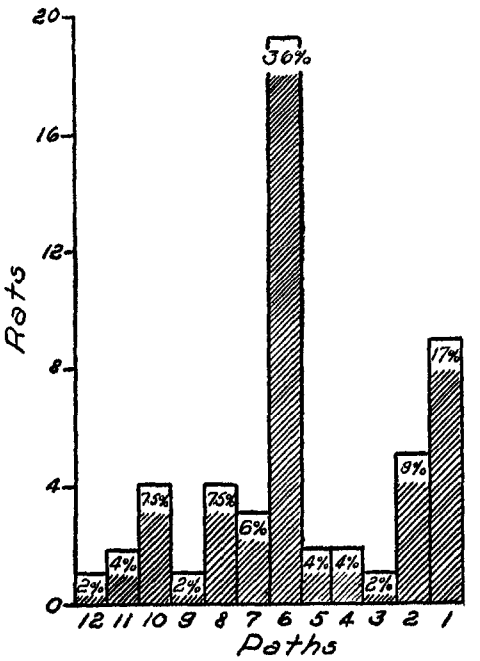


FIG. 5. NUMBER OF RATS WHICH CHOSE EACH OF THE PATHS

intensity of the light (Group B) were instrumental in the performance of the animals on the test trial. One animal in Group B chose path 6, the path which pointed directly to the location of the food box during preliminary training.

EXPERIMENT II

Tolman, Ritchie, and Kalish raise the question in their study as to why all animals did not choose the short-cut path, path 6. They propose the following hypothesis:

> We believe that the reason that the remaining rats failed to take the short-cut was that they had not had enough training and thus had not yet learned the location of the food. With a few more days of training, we should have expected that the remaining rats would have chosen the short-cut (pp. 21–22).

The present experiment was designed to investigate this hypothesis experimentally.

Animals. Twenty-two animals were used in this experiment and none had had previous maze experience. They were approximately 90 days old at the beginning of the experiment. Twenty-one animals made a choice on the test trial within six minutes.

Apparatus. The apparatus used for preliminary training and for the test trial was the same as that used in Experiment I. The light was at H_1 for all animals throughout the study.

Pre-test procedures. Preliminary feeding and training was the same as for the animals in Experiment I for the first four days.

On Day 5, the animals were given three trials. On each trial, each animal was placed at A and allowed to explore the tabletop until it had found the entrance to alley CD and had proceeded to the food box.

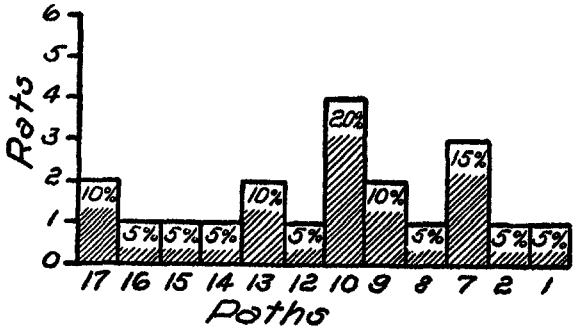


FIG. 6. NUMBER OF CHOICES WITH NINE DAYS' TRAINING

On Day 6, the animals were given three trials under the same conditions as for Day 5.

On Days 7, 8, and 9, the animals were given one trial a day under the same conditions as for Days 5 and 6.

On Day 9, all animals went immediately to the entrance of alley CD, entered and proceeded to the food box.

Test procedures. On the 10th day, one test trial was given. The procedures and the criterion of a choice were the same as for Group A in Experiment I. The light was at H_1.

Results of experiment II. We present the results on the test trial in figure 6. These data show that no one path received a very large percentage of choices. The results show a grouping around paths 7, 8, 9, and 10 (50 per cent), with 30 per cent of the choices on paths 9 and 10, the paths adjacent to alley CD.

EXPERIMENT III

The results of Groups A (figure 3), B, and C (figure 4) in Experiment I and Experiment II (figure 6) show a tendency for animals to choose paths close to alley CD. We assume that (1) the presence of alley CD on the test trial was in

part responsible for this result, or (2) that a forward going response-tendency developed as a result of preliminary training. We decided to run the test trial without alley CD.

Animals. Twenty-seven animals about 120 days old were used. The animals had had no previous maze experience. All animals made a choice on the test trial.

Apparatus. The same apparatus used for Group A in Experiment I was used.

Pre-test procedures. The pre-test training was the same as for Group A in Experiment I.

Test procedures. The test trial was given on the 5th night. Alley CD was removed from the apparatus shown in figure 2. The same criterion of a choice was used as that used in Experiment I.

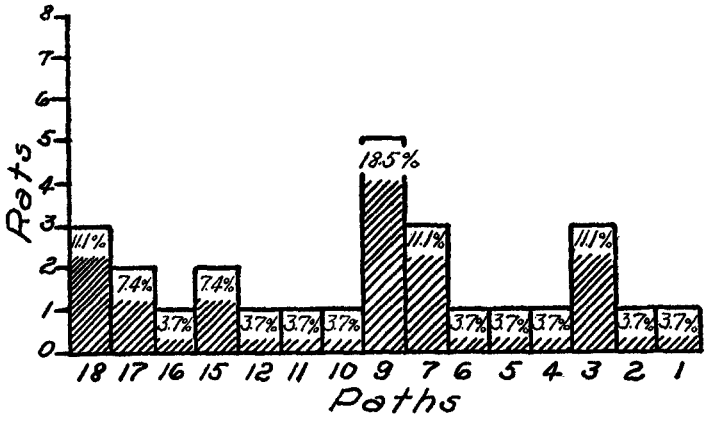


FIG. 7. NUMBER OF CHOICES WITH ALLEY CD REMOVED

Results of experiment III. We present the results of this experiment in figure 7. These data show only a slight tendency for animals to choose paths 9 and 10, the paths adjacent to the former position of alley CD. Twenty-two and two tenths of the animals chose paths 9 and 10 with CD removed. In Experiment I, 37.5 per cent of Group A and 38 per cent of Group B chose paths 9 and 10 with CD present. This supports our assumption about the function of alley CD on the test trial in Experiment I and II. Some of the choices adjacent to CD in Experiments I and II might be a function of an artifact of the experimental apparatus.

EXPERIMENT IV

The results of Group A in Experiment I (figure 3) show not only a tendency for the animals to choose paths 9 and 10, paths adjacent to alley CD, but also a stronger tendency for animals to choose paths 11 through 18 (45.8 per cent) than paths 1 through 8 (16.7 per cent). We thought that this result might be a function of the running of path DE during preliminary training (see figure 1). We decided to increase the length of path DE 42 inches, making it 60 inches, and

also path FG 42 inches in order to leave the food box in the same position as in Experiment I. We assumed that this change in the length of DE might increase the tendency to choose paths to the left of alley CD if the increase was not nullified by the same increase in the length of FG.

Animals. Eighteen animals about 120 days old were used in this study. None had had previous maze experience. One failed to make a choice within six minutes on the test trial.

Apparatus. The apparatus used in preliminary training was the same as that used in Experiment I (figure 1), but with DE increased to 60 inches and FG to 102 inches. On the test trial the apparatus used was the same as for Experiment I. A light was at H_1 during both the preliminary training and the test trial.

Pre-test procedures. The procedures used were the same as those for Group A in Experiment I.

Test-procedures. The procedures used were the same as those used for Group A in Experiment I.

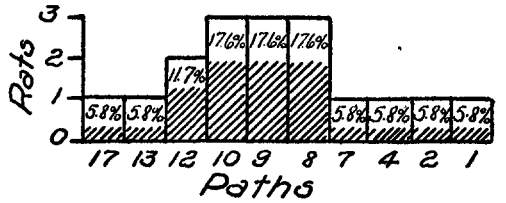


FIG. 8. NUMBER OF CHOICES FOLLOWING MODIFIED PRE-TEST TRAINING

Results of Experiment IV. We present the results for Experiment IV in figure 8. These data show no increased tendency to choose paths 11 to 18 over the results of Group A in Experiment I. However, there is still a tendency to choose paths 9 and 10 (35.2 per cent), the pathways adjacent to alley CD.

EXPERIMENT V

The results of Group A in Experiment I (figure 3) show a tendency (37.5 per cent) for animals to choose paths adjacent to alley CD. The animals in this experiment had four days of preliminary training before the test trial. In Experiment II, the experiment in which the animals received nine days of preliminary training, the animals chose paths adjacent to alley CD more frequently (30 per cent) than other paths. When alley CD was removed on the test trial in Experiment III, the tendency to choose paths adjacent to the former position of CD was reduced (22.2 per cent). In Experiment IV, 35.2 per cent of the animals chose paths 9 and 10, with CD present on the test trial, following an increase in the length of path DE and FG during the preliminary training period.

In the present experiment we decided to give the test trial without preliminary training.

Animals. One hundred and three animals were used. The animals were approximately 100 days old and had had no previous maze experience. They

were divided into two groups. Group A ran the test trial with alley CD in position. Group B ran the test trial with alley CD removed from the apparatus shown in figure 2.

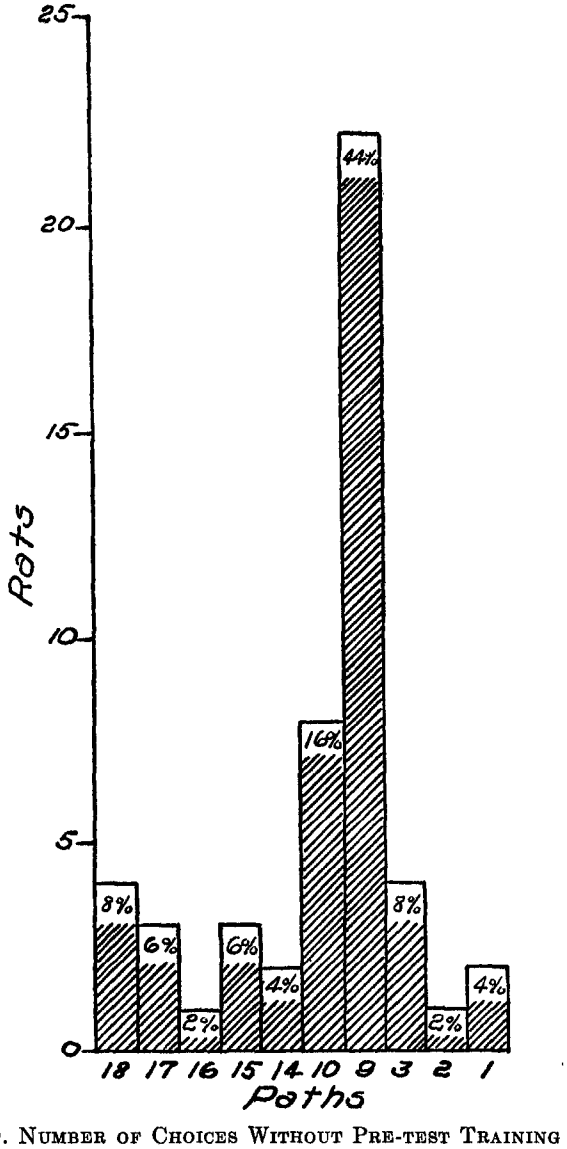


FIG. 9. NUMBER OF CHOICES WITHOUT PRE-TEST TRAINING

Apparatus. The 60 animals in Group A were given the test trial on the apparatus used for the test trial of Group A in Experiment I. Fifty animals made a choice within six minutes. The 43 animals in Group B were given the test trial on the apparatus used for the test trial in Experiment II. Alley CD was removed. Forty animals made a choice within six minutes.

Test procedures. The animals were not fed for 48 hours prior to the test trial. The animals in Group A were not required to enter alley CD before making a choice. If the animal did not choose a path and run to the end of it within six minutes, it was removed from the experiment.

Results of experiment V. We present the results of the test trial for Group A in figure 9. These data show a marked tendency (60 per cent) for the animals to choose paths 9 and 10, paths adjacent to alley CD.

We present the results of the test trial for Group B in figure 10. These data show no tendency (10 per cent) for the animals to choose paths near the former position of alley CD, but rather a tendency to choose paths 1 and 2 (25 per cent) and 15–18 (37.5 per cent). The animals had a tendency to choose the terminal paths. The choices adjacent to CD for Group A (60 per cent) with alley CD

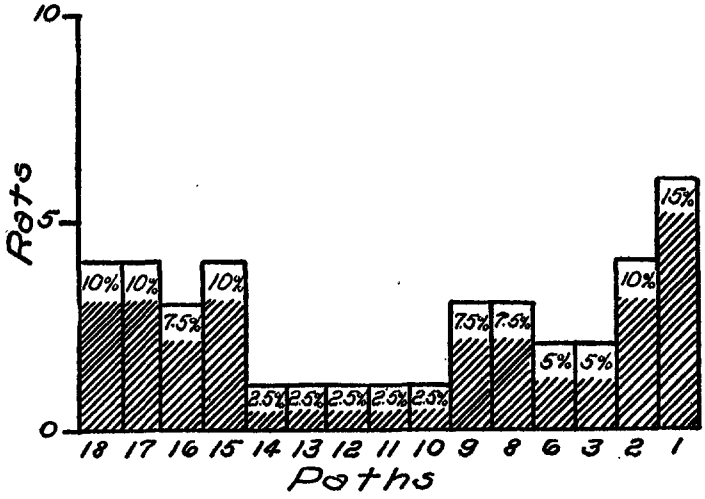


FIG. 10. NUMBER OF CHOICES WITHOUT PRE-TEST TRAINING; CD REMOVED

in the apparatus on the test trial, and the absence of a tendency to choose paths 9 and 10 in Group B with alley CD removed (10 per cent), led us to assume that the results are a function of an artifact of the experimental set-up. The choices of terminal paths by Group B also appear to be a function of an artifact of the experimental set-up (cf. Tolman, Ritchie, and Kalish, p. 18).

DISCUSSION OF RESULTS

It is evident from the results presented in this study that different positions of the light during preliminary training and on the test trial did not influence the performance of the animals on the test trial. The animals did not choose the short-cut path (path 6) to the former location of the food box, which was indicated by the position of the light for Groups A and B in Experiment I on the test trial. The animals not only did not choose path 6, the shortcut path, on the test trial, but they exhibited no disposition to choose paths adjacent to path 6. Differences in the intensity of the light for Groups A (7.5-watt) and B (40-watt)

at H_1 did not affect the performance. Thirty-seven and one-half per cent of the animals in Group A and 38 per cent of those in Group B chose paths 9 and 10, paths adjacent to alley CD, and one animal of Group B chose path 6, the path which pointed directly to the location of the food box during preliminary training. When the light was at H_2 during preliminary training and on the test trial, the results were substantially the same as those in Group A and B, 32.2 per cent choosing paths 9 and 10.

These results are at variance with those reported by Tolman, Ritchie, and Kalish, in which 36 per cent of the animals chose path 6, the short-cut path, and 9.5 per cent paths 9 and 10. Since we found no differences in performance between A and B in Experiment I, when a 7.5-watt and a 40-watt light was used, we do not feel that the 5-watt light used by Tolman, Ritchie, and Kalish will account for this difference.

Tolman, Titchie, and Kalish attribute their failure to get 100 per cent choices on the short-cut to the fact that the animals had not "had enough training and thus had not yet learned the location of the food. With a few more days of training we should have expected that the remaining rats would have chosen the short-cut (pp. 21–22)". We gave the animals in Experiment II nine days of preliminary training, and yet failed to get any evidence of a disposition on the part of the animals to choose the short-cut. In fact, we obtained the same type of clustering around CD which we found in Experiment I.

The grouping of choices around alley CD was observed in Groups A (37.5 per cent), B (38 per cent), and C (32.2 per cent) in Experiment I and in Experiment II (30 per cent). We assume that the presence of CD on the test trial contributed to this result. In Experiment III the test trial was run without CD. The results here show that only 22.2 per cent of the animals chose paths 9 and 10, paths adjacent to the former position of CD. This result might be a function of a response-tendency developed during preliminary training, and a function of the gap in the apparatus on the test trial resulting from the removal of CD.

The results of Group A in Experiment I (figure 3) show a stronger tendency for animals to choose paths 11 to 18 (45.8 per cent) than paths 1 to 8 (16.7 per cent). We thought that this result was a function of the animals' running path DE during preliminary training. In Experiment IV path DE was lengthened 42″ and path FG was lengthened by the same amount in order that the food box would remain in the same position at H_1 during preliminary training. This change in the apparatus did not develop any increased tendency to choose paths 11 to 18 over the results of Groups A in Experiment I. However, the animals did choose paths 9 and 10 (35.2 per cent) more frequently than other paths.

With alley CD present on the test trial we observed a consistent disposition on the part of the animals to choose paths 9 and 10 regardless of the position of the light, the intensity of the light, an increase in the length of paths DE and FG, or with five or nine days of training. In Experiment V, the experiment in which the animals were given the test trial without preliminary training, we found that 60 per cent of the animals of Group A chose path 9 and 10 when CD was

present and that 10 per cent of the animals of Group B chose path 9 and 10 when CD was absent.

SUMMARY AND CONCLUSIONS

We have failed to confirm the results of Tolman, Ritchie, and Kalish in four experiments in which preliminary training was given. In Experiment V, an experiment in which no preliminary training was given, we got results similar to the results obtained in those experiments in which preliminary training was given. Either our animals did not learn to expect food at location L, or if they did, they did not have the capacity to determine the short cut. We are led to explain our results, in part at least, in terms of the presence or absence of CD on the test trial. Moreover, we are led to question the explanation given by Tolman, Ritchie, and Kalish of their results. Their explanation appears to us to involve two assumptions which do not mutually imply each other. One is that each of their correct choice rats did in fact learn the location of the food box as a result of the training. While we found no evidence, even in the case of the group given 9 days of training—five days more training than Tolman, Ritchie, and Kalish gave their rats—we would argue that even if their rats did learn the locus of the food box, such learning is in itself inadequate to explain their 100 per cent correct choices. In view of the close proximity of paths 5 and 7 to path 6 (see figure 2) and the absence of distinguishable features at the choice points in the apparatus, it seems very dubious to us that the 100 per cent correct choice of path 6 is explicable simply in terms of the assumption that the correct choice rats had learned the location of the food box. Under such conditions the best that one could expect would be a clustering of choices around path 6.

It appears to us, then, that the Tolman, Ritchie, and Kalish explanation involves a second presupposition. This assumption is that their rats possessed a capacity to discriminate between the 18 novel available pathways, without distinguishable features at the choice points, with which they were confronted in the test trial, and to determine (infer) without error that No. 6 pathway was a short-cut, *i.e.*, the pathway to location L. We question the legitimacy of the assumption that rats have such a high order capacity, and hence the legitimacy of a theory which presupposes that they do. We certainly found no evidence that any of the rats that we used had such a capacity and much evidence that none of them did. If rats have such a capacity and if rats do, when subjected to the kind of training employed, learn the spatial locus of the food box (develop an expectation of food at location L), it appears to us that we would have got numbers of choices on the short-cut at least greater than chance. Certainly we should have found evidence of this capacity in some of the rats used in our study, especially in those subjected to five days more training (Experiment II), if any rat, when subjected to the kind of training that we and Tolman, Ritchie, and Kalish employed, does in fact learn the locus of the food box (develops an expectation of food at location L).

Consequently, since our results for both the animals with training and without

training are in part explainable in terms of an artifact of the experimental apparatus (the presence of alley CD), we wonder if their results are not explainable in terms of some artifact of their experimental set-up, and hence if there is any need for them to appeal to the senior author's expectancy theory in its inferential form.

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